US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250287605
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	HUANG; Yen-Chieh et al.

SEMICONDUCTOR DEVICE WITH LOW CURRENT LEAKAGE AND METHOD FOR MANUFACTURING THE SAME

Abstract

A method for manufacturing a semiconductor structure includes: forming a gate electrode; forming a gate dielectric layer over the gate electrode; forming a channel over the gate dielectric layer opposite to the gate electrode, the channel including a semiconductor material; forming a first confinement layer on the channel opposite to the gate electrode, an energy band gap of the first confinement layer being greater than an energy band gap of the channel; and forming a drain electrode and a source electrode which are connected to the channel, the drain electrode and the source electrode being spaced apart from each other and each including an electrically conductive material.

Inventors: **HUANG; Yen-Chieh (Hsinchu, TW), HUANG; Huai-Ying (Hsinchu, TW), SHIH; Yu-Chuan (Hsinchu, TW), LU; Chun-Chieh (Hsinchu, TW), LEE; I-Che (Hsinchu, TW), LIN; Yu-Ming (Hsinchu, TW)**

Applicant: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.** (Hsinchu, TW)

Family ID: **96949882**

Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD. (Hsinchu, TW)**

Appl. No.: **18/600934**

Filed: **March 11, 2024**

Publication Classification

Int. Cl.: **H10B51/30** (20230101); **H01L21/28** (20250101); **H01L29/417** (20060101); **H01L29/51** (20060101); **H01L29/66** (20060101); **H01L29/78** (20060101); **H01L29/786** (20060101)

U.S. Cl.:

CPC **H10B51/30** (20230201); **H10D30/031** (20250101); **H10D30/0415** (20250101); **H10D30/6729** (20250101); **H10D30/6755** (20250101); **H10D30/6757** (20250101); **H10D30/701** (20250101); **H10D64/033** (20250101); **H10D64/689** (20250101);

Background/Summary

BACKGROUND

[0001] Thin film transistors, which are manufactured using thin film techniques, are known to be used in various applications. The thin film transistors may be formed in a front-end-of-line process, or may be embedded in a back-end-of-line interconnect structure, thereby reducing chip area of an integrated circuit. The thin-film transistors with low current leakage are under continuous development.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. **1** is a schematic sectional view illustrating a semiconductor structure including a semiconductor device in accordance with some embodiments.

[0004] FIGS. **2** and **3** are each a schematic sectional view illustrating the semiconductor device in accordance with some other embodiments.

[0005] FIG. **4** is a flow diagram illustrating a method for manufacturing the semiconductor device in accordance with some embodiments.

[0006] FIGS. **5** to **11** illustrate schematic views of intermediate stages of the method depicted in FIG. **4** in accordance with some embodiments.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “on,” “above,” “top,” “bottom,” “upper,” “lower,” “over,” “beneath,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing amounts, sizes, dimensions, proportions, shapes, formulations, parameters, percentages, quantities, characteristics, or other numerical values used in the specification and claims, are to be understood as being modified in all instances by the terms “about” and “substantially” even if the terms “about” and “substantially” are not explicitly recited with the values, amounts or ranges. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and appended claims are not and need not be exact, but may be approximations and/or larger or smaller than specified as desired, may encompass tolerances, conversion factors, rounding off, measurement error, and other factors known to those of skill in the art depending on the desired properties sought to be obtained by the presently disclosed subject matter. For example, the terms “about” and “substantially,” when used with a value, can capture variations of, in some aspects ±10%, in some aspects ±5%, in some aspects ±2.5%, in some aspects ±1%, in some aspects ±0.5%, and in some aspects ±0.1% from the specified amount, as such variations are appropriate to perform the disclosed methods or employ the disclosed compositions.

[0010] The term “source/drain portion(s)” may refer to a source or a drain, individually or collectively dependent upon the context.

[0011] In some applications, a gate dielectric layer of a thin-film transistor (TFT) may be made of a ferroelectric material, and such thin-film transistor (also referred to as a field-effect transistor (FeFET)) may function as a memory device capable of storing binary data. In the case that the FeFET includes an n-channel (i.e., a channel having an n-type conductivity), the threshold voltage of the FeFET (in an initial state, a programmed state and/or an erased state) may be a negative value, indicating that a current (i.e., current leakage) is present in the n-channel when no voltage is applied. Therefore, in order to reduce the current leakage, the present disclosure provides a semiconductor device having a relatively positive threshold voltage when an n-channel is included therein, or a semiconductor device having a relatively negative threshold voltage when a p-channel (i.e., a channel having a p-type conductivity) is included therein.

[0012] FIG. **1** is a schematic sectional view illustrating a semiconductor structure **1** in accordance with some embodiments. The semiconductor structure **1** includes a substrate **10**, a first semiconductor device **20** (which serves as a front-end-of-line (FEOL) transistor), an inter-layer dielectric (ILD) layer **11** formed on the substrate **10** to cover the first semiconductor device **10**, an interconnect structure **30** formed on the ILD layer **11**, and a second semiconductor device **40** (which serves as a back-end-of-line (BEOL) transistor) formed in the interconnect structure **30**. It is noted that the second semiconductor device **40** is not limited to be located directly above the first semiconductor device **20**. In some other embodiments, the first semiconductor device **20** shown in FIG. **1** may be omitted. In such case, the second semiconductor device **40** may be directly formed on the substrate **10**, or formed on a buffer layer (not shown) that is preformed on the substrate **10** and that is used for improving the film quality of a film to be formed thereon.

[0013] In some embodiments, the substrate **10** may include elemental semiconductor materials (such as crystalline silicon, diamond, or germanium), compound semiconductor materials (such as silicon carbide, gallium arsenide, indium arsenide, or indium phosphide), alloy semiconductor materials (such as silicon germanium, silicon germanium carbide, gallium arsenide phosphide, or gallium indium phosphide), or combinations thereof. In some embodiments, the substrate **10** may be a bulk semiconductor substrate, for example, but not limited to, a bulk substrate of silicon, germanium, silicon germanium, or other suitable semiconductor materials (such as the examples described earlier in the same paragraph). In some other embodiments not shown herein, the substrate **10** may be configured as a semiconductor-on-insulator substrate. In some embodiments, the semiconductor material in the substrate **10** may be un-doped, or may be doped with impurities (e.g., n-type impurities or p-type impurities) to form a well portion for the first semiconductor device **20**. In some yet other embodiments, the substrate **10** may be a glass substrate. Other suitable materials and configurations for the substrate **10** are within the contemplated scope of the present disclosure.

[0014] In some embodiments, the first semiconductor device **20** may be a field-effect transistor (FET), and includes a channel **21**, two source/drain portions **22** formed at two opposite sides of the channel **21**, a gate dielectric layer **23** formed on the channel **21**, a gate electrode **24** formed on the gate dielectric layer **23** such that the channel **21** is spaced apart from the gate electrode **24** by the gate dielectric layer **23**, and two dielectric spacers **25** formed at two opposite sides of the gate electrode **24**. In some embodiments, the first semiconductor device **20** further includes two source/drain contacts **26** and a gate contact **27** formed in the ILD layer **11** and spaced apart from each other. The source/drain contacts **26** are respectively formed on the source/drain portions **22**, and the gate contact **27** is formed on the gate electrode **24**. In some embodiments, as shown in FIG. **1**, the first semiconductor device **20** may be configured as a planar FET, in which (i) the source/drain portions **22** are formed in the substrate **10** by an implantation process, and (ii) a portion of the substrate **10**, which is located between the source/drain portions **22**, serves as the channel **21**. The source/drain portions **22** may be doped with impurities to have an n-type conductivity or a p-type conductivity according to the type of the first semiconductor device **20** (i.e., the source/drain portions **22** have the n-type conductivity when the first semiconductor device **20** is an n-FET; and the source/drain portions **22** have the p-type conductivity when the first semiconductor device **20** is a p-FET). In some embodiments, the gate dielectric layer **23** may be made of silicon oxide, and the gate electrode **24** may be made of polycrystalline silicon. In some other embodiments not shown herein, the first semiconductor device **20** may be configured as a fin-type field-effect transistor (FinFET), or a gate-all-around field-effect transistor (GAAFET). In such case, the gate dielectric layer **23** may include a high dielectric constant (high-k) material, and the gate electrode **24** include a metallic material. Other three-dimensional (3D) transistor structures suitable for the first semiconductor device **20** are within the contemplated scope of the present disclosure.

[0015] In some embodiments, the ILD layer **11** includes a dielectric material. In some embodiments, the dielectric material for forming the ILD layer **11** may have a low dielectric constant, and may include silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), silicon oxycarbide (SiOC), spin-on-glass (SOG), or combinations thereof. Other dielectric materials suitable for the ILD layer **11** are within the contemplated scope of the present disclosure. The interconnect structure **30** includes a plurality of interconnect layers which are sequentially formed on the ILD layer **11**. Four of the interconnect layers are exemplarily shown in FIG. **1**, and are respectively represented by M.sub.0, M.sub.x−1, M.sub.x, M.sub.x+1, where x is an integer not less than 2. Other interconnect layers between the interconnect layers M.sub.0 and M.sub.x−1 are omitted. Each of the interconnect layers M.sub.0, ... M.sub.x−1, M.sub.x, M.sub.x+1 includes an inter-metal dielectric (IMD) portion **31**, and a plurality of electrically conductive elements **32** (for example, metal contacts, metal lines, and/or metal vias) formed in the IMD portion **31**. Each of the electrically conductive elements **32** in each of the interconnect layers M.sub.0, ... M.sub.x−1, M.sub.x, M.sub.x+1 is connected to a corresponding one of the electrically conductive elements **32** in an adjacent one of the interconnect layers M.sub.0, ... M.sub.x−1, M.sub.x, M.sub.x+1. In some embodiments, the second semiconductor device **40** is formed in the IMD portion **31** of the interconnect layer M.sub.x. In some embodiments, the electrically conductive elements **32** may include a low resistance electrically conductive material such as copper (Cu), cobalt (Co), ruthenium (Ru), molybdenum (Mo), chromium (Cr), tungsten (W), manganese (Mn), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), golden (Au), silver (Ag), aluminum (Al), osmium (Os), alloys thereof, or combinations thereof. Possible low dielectric constant (low-k) materials suitable for the IMD portion **31** are similar to those for forming the ILD layer **11**, and thus the details thereof are omitted for the sake of brevity.

[0016] In some embodiments, the second semiconductor device **40** may be a thin-film transistor (TFT), and includes a channel **41** that includes a semiconductor material, a gate dielectric layer **42** formed on the channel **41**, a gate electrode **43** formed on the gate dielectric layer **42** such that the channel **41** is separated from the gate electrode **43** by the gate dielectric layer **42**, a drain contact unit **51** and a source contact unit **52** connected to the channel **41** and spaced apart from each other, and a first confinement layer **44** disposed on the channel **41** opposite to the gate electrode **43**.
[0017] In some embodiments, as shown in FIG. **1**, the gate electrode **43** is located beneath the channel **41** such that the gate electrode **43** and the channel **41** are respectively located proximate to and distal from the substrate **10**. In such case, the second semiconductor device **40** is referred to as a bottom-gate TFT, but is not limited thereto. In some other embodiments not shown herein, the second semiconductor device **40** may be configured as a top-gate TFT, a double-gate TFT, a vertical TFT, or 3-dimensional TFT. For the top-gate TFT, the gate electrode and the channel are respectively located distal from and proximate to the substrate. The double gate TFT includes a lower gate electrode and an upper gate electrode which are respectively separated from the channel by a lower gate electric layer and an upper gate dielectric layer. The lower and upper gate electrodes are disposed at two opposite sides of the channel, and are respectively located proximate to and distal from the substrate. In some embodiments, the second semiconductor device **40** may be applied to static random-access memory (SRAM), dynamic random-access memory (DRAM, e.g., one-transistor/one-capacitor (1T-1C) DRAM cell, 3-dimensional DRAM structure, standalone DRAM structure, embedded DRAM structure, etc.), ferroelectric memories (e.g., 1T-1C ferroelectric random-access memory memory (1T-1C FeRAM), 1T FeRAM, metal-ferroelectric-metal field-effect transistor-based (MFMFET-based) FeRAM, metal-ferroelectric-metal-insulator-semiconductor field-effect transistor-based (MFMISFET) FeRAM, etc.), or peripheral devices (e.g., power gates, input/output (I/O) devices, or selectors for memory cells, etc.). It is noted that the second semiconductor device **40** is not limited to be formed in the IMD portion **31** of the interconnect layer M.sub.x. In some embodiments not shown herein, the second semiconductor device **40** may be formed in any one of the interconnect layers M.sub.0, ... M.sub.x−1, M.sub.x+1.
[0018] In some embodiments, as shown in FIG. **1**, the gate electrode **43** is formed on the IMD portion **31** of the interconnect layer M.sub.x−1, and is connected to one of the electrically conductive elements **32** of the interconnect layer M.sub.x−1. In some embodiments, the gate electrode **43** includes tungsten (W), platinum (Pt), aluminum (Al), silver (Ag), copper (Cu), nickel (Ni), or alloys thereof. Other metallic material suitable for forming the gate electrode **43** are within the contemplated scope of the present disclosure.
[0019] In some embodiments, the gate dielectric layer **42** is formed on the gate electrode **43** (in other words, the gate electrode **43** is formed on a lower surface of the gate dielectric layer **42**). In some embodiments, the gate dielectric layer **42** may include silicon oxide, silicon nitride, silicon oxynitride, or a high-k dielectric material (such as hafnium oxide, hafnium tantalum oxide, zirconium oxide, zirconium aluminum oxide, hafnium aluminum oxide, hafnium silicon oxide, aluminum oxide, etc.). In some other embodiments, when the second semiconductor device **40** is formed as a FeRAM, the gate dielectric layer **42** may include a ferroelectric material (such as hafnium zirconium oxide (HZO), barium titanate (BaTiO.sub.3), lead titanate, lead zirconate, lithium niobate, sodium niobate, potassium niobate, potassium tantalite, bismuth scandate, bismuth ferrite, aluminum scandium nitride, or hafnium oxide doped with yttrium (Y), lanthanum (La), gadolinium (Gd), erbium (Er), titanium (Ti), zirconium (Zr), aluminum (Al), or tantalum (Ta)). Other suitable materials for the gate dielectric layer **42** are within the contemplated scope of disclosure. In some embodiments, the gate dielectric layer **42** has a thickness ranging from about 1 nm to about 500 nm. In some embodiments, as shown in FIG. **1**, the gate dielectric layer **42** has a first region **421** for forming the channel **41** thereon, two second regions **422** respectively for forming the drain contact unit **51** and the source contact unit **52** thereon, and a third region **423** for

being covered by the IMD portion **31** of the interconnect layer M.sub.x. The regions **421**, **422**, **423** are displaced from each other.

[0020] In some embodiments, the channel **41** is formed on the first portion **421** of the gate dielectric layer **42** (in other words, the gate dielectric layer **42** is formed on a lower surface of the channel **41**). In some embodiments, the semiconductor material for forming the channel **41** includes silicon (Si), germanium (Ge), silicon germanium, silicon germanium carbide, gallium arsenic, indium phosphide, gallium phosphide, gallium nitride, gallium antimony, aluminum arsenic, indium arsenic, indium antimony, aluminum gallium arsenic, gallium indium arsenic, gallium indium phosphide, indium aluminum arsenic, aluminum indium gallium phosphide, cadmium sulfide, cadmium selenide, zinc sulfide, zinc selenide, zinc telluride, lead sulfide, lead telluride, mercury telluride, or combinations thereof. In some other embodiments, the semiconductor material for forming the channel **41** may have an n-type conductivity. That is, the channel **41** may include an n-type metal oxide semiconductor, such as indium gallium zinc oxide (IGZO), indium zinc oxide (IZO), indium gallium tin oxide (IGSnO), indium gallium tin zinc oxide (IGSnZnO), which has extra charge carriers (i.e., electrons) capable of moving in the channel **41** during operation of the second semiconductor device **40**. In such case, the second semiconductor device **40** using the n-type metal oxide semiconductor as the channel **41** is referred to as an n-type TFT, electrons are majority carriers in the channel **41**, and such channel **41** may be referred to as an n-channel. In some embodiments, when the channel **41** is made of IGZO, such IGZO may be represented by a chemical formula of (Ga.sub.2).sub.x1(In.sub.2).sub.(1-x1)Zn.sub.y1O.sub.7, where x1 ranges from about 0.2 to about 0.75, and y1 is greater than zero and not greater than about 0.75. In some yet other embodiments, the semiconductor material for forming the channel **41** may have a p-type conductivity. That is, the channel **41** may include a p-type metal oxide semiconductor, such as copper oxide (Cu.sub.2O), tin oxide (SnO), nickel oxide (NiO), K.sub.2Sn.sub.2O.sub.3, Al-doped NiO (Al:NiO), V-doped NiO (V:NiO), Cu-doped NiO (Cu:NiO), Sn-doped NiO (Sn:NiO), Mg-doped NiO (Mg:NiO), Li-doped MgNiO (Li:MgNiO), RbSn.sub.2O.sub.3, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2Sn.sub.2O.sub.6, Pb-doped CsSnI.sub.3 (Pb:CsSnI.sub.3), CuCrO.sub.2, Mg-doped Cu.sub.2O (Mg:Cu.sub.2O), CuFeO.sub.2, CsPbIBr.sub.2, or phenyl-C61-butyric acid methyl ester (PCBM), which has extra charge carriers (i.e., holes) capable of moving in the channel **41** during operation of the second semiconductor device **40**. In such case, the second semiconductor device **40** using the p-type metal oxide semiconductor as the channel **41** is referred to as a p-type TFT, holes are majority carriers in the channel **41**, and such channel **41** may be referred to as a p-channel. In some embodiments, the channel **41** may have a thickness ranging from about 1 nm to about 100 nm.

[0021] In some embodiments, the drain contact unit **51** and the source contact unit **52** are respectively disposed on the second regions **422** of the gate dielectric layer **42**, and are respectively located at two opposite sides of the channel **41** (in other words, the drain contact unit **51** and the source contact unit **52** are respectively connected to a drain-side surface **411** and a source-side surface **412** of the channel **41**). In some embodiments in which the second semiconductor device **40** is formed as a TFT, when a specified voltage (V.sub.GS) is applied between the gate electrode **43** and the source contact unit **52** to bias the second semiconductor device **40** to an on-state, the majority carriers (electrons in an n-channel or holes in a p-channel) may flow in a direction from the source contact unit **52** toward the drain contact unit **51**, and thus an on-state drain current (I.sub.on) may be detected at the drain contact unit **51**. In some embodiments, the on-state drain current (I.sub.on) may increase as the thickness of the channel **41** increases. For example, the on-state drain current (I.sub.on) may increase from about 60 μA/μm to about 200 μA/μm as the thickness of the channel **41** increases from about 1 nm to about 100 nm. In some other embodiments, when the second semiconductor device **40** serves as a memory device (e.g., a FeRAM), and includes the drain and source contact units **51**, **52** having the arrangement as described above (i.e., the drain and source contact units **51**, **52** are in direct contact with the gate dielectric layer **42**), a polarizable region of the gate dielectric layer **42** can be enlarged, and a memory window of the FeRAM may be enlarged accordingly. Furthermore, when a read voltage (V.sub.GS) is applied between the gate electrode **43** and the source contact unit **52**, the majority carriers (electrons in an n-channel or holes in a p-channel) may flow in a direction from the source contact unit **52** toward the drain contact unit **51**, and thus a programmed state or an erased state of the FeRAM may be determined according to the on-state drain current (I.sub.on) detected at the drain contact unit **51**.

[0022] In some embodiments, as shown in FIG. **1**, the drain contact unit **51** includes a drain-side barrier layer **511** connected to the drain-side surface **411** of the channel **41**, a drain-side conductive metal oxide layer **512** formed on the drain-side barrier layer **511**, a drain-side diffusion blocking layer **513** formed on the drain-side conductive metal oxide layer **512**, and a drain electrode **514** formed on the drain-side diffusion blocking layer **513**. The source contact unit **52** includes a source-side barrier layer **521** connected to the second-side surface **412** of the channel **41**, a source-side conductive metal oxide layer **522** formed on the source-side barrier layer **521**, a source-side diffusion blocking layer **523** formed on the source-side conductive metal oxide layer **522**, and a source electrode **524** formed on the source-side diffusion blocking layer **523**.

[0023] The drain-side barrier layer **511** is provided to form a first barrier height at a first interface between the drain-side barrier layer **511** and the channel **41**, and the source-side barrier layer **521** is provided to form a second barrier height at a second interface between the source-side barrier layer **521** and the channel **41**. The first barrier height and the second barrier height may vary depending on the types of materials selected for the drain-side barrier layer **511** and the source-side barrier layer **521**, and thus a threshold voltage of the second semiconductor device **40** may be varied. In some embodiments in which the semiconductor device **40** serves as a memory device, the threshold voltage of the semiconductor device **40** may be an initial threshold voltage (i.e., the threshold voltage when the memory device **40** is in an initial state, V.sub.t(initial)), a programmed threshold voltage (i.e., the threshold voltage when the memory device **40** is in a programmed state, V.sub.t(PRG)), or an erased threshold voltage (i.e., the threshold voltage when the memory device **40** is in an erased state, V.sub.t(ERS)). In some embodiments, an energy band gap of the drain-side barrier layer **511** is greater than an energy band gap of the channel **41**. In some embodiments, an energy band gap of the source-side barrier layer **521** is greater than the energy band gap of the channel **41**.

[0024] In some embodiments in which the second semiconductor device **40** is an n-type TFT and the channel **41** is an n-channel (i.e., the semiconductor material for forming the channel **41** has an n-type conductivity), with the provision of the drain-side and source-side barrier layers **511**, **521** (each having a conduction band edge energy that is higher than a conduction band edge energy of the n-channel **41**), the threshold voltage of the n-type TFT **40** can be shifted from a relatively negative value (without the drain-side and source-side barrier layers **511**, **521**) to a relatively positive value (a negative value with a smaller absolute value or a positive value). As such, when no voltage is applied to the semiconductor device **40** or the semiconductor device **40** is in a standby mode, a static power dissipation, which results from leakage current in the semiconductor device **40**, may be suppressed. The first barrier height formed at the first interface is the difference between the conduction band edge energy of the n-channel **41** and the conduction band edge energy of the drain-side barrier layer **511**, and may be referred to as a first electron barrier height. The second barrier height formed at the second interface is the difference between the conduction band edge energy of the n-channel **41** and the conduction band edge energy of the source-side barrier layer **521**, and may be referred to as a second electron barrier height. In some embodiments, the first electron barrier height is greater than the second electron barrier height, and thus degradation of an on-state drain current may be prevented or alleviated (i.e., the on-state drain current (I.sub.on) of the second semiconductor device **40** is relatively large).

[0025] In some embodiments in which the second semiconductor device **40** is an n-type TFT, the

drain-side barrier layer **511** includes a drain barrier material, and the source-side barrier layer **521** includes a source barrier material. Each of the drain barrier material and source barrier material may be independently selected from gallium oxide, silicon oxide, or aluminum oxide. In some embodiments, each of the drain-side barrier layer **511** and the source-side barrier layer **521** may further include dopants so as to adjust the energy band gap of the drain-side barrier layer **511** (or the source-side barrier layer **521**). In some embodiments, first dopants doped in the drain barrier material include indium, zinc or a combination thereof. In some embodiments, second dopants doped in the source barrier material include indium, zinc or a combination thereof. For example, each of the drain-side barrier layer **511** and the source-side barrier layer **521**, which may be made of gallium oxide, zinc-doped gallium oxide or indium-doped gallium oxide, may be represented by a chemical formula of (Ga.sub.x1A.sub.(1-x1))O.sub.y1, where A is Zn or In, x1 ranges from about 0.1 to about 1, and y1 ranges from about 0.1 to about 3. As a concentration of the first dopants in the drain-side barrier layer **511** (or a concentration of the second dopants in the source-side barrier layer **521**) increases, the energy band gap of the drain-side barrier layer **511** (or the source-side barrier layer **521**) decreases. The greater the energy band gap of each of the drain-side and source-side barrier layers **511**, **521** is, the greater the shift of the threshold voltage of the n-type TFT **40** is. In other words, with the provision of the drain-side and source-side barrier layers **511**, **521** each having a relatively great energy band gap, the n-type TFT **40** may have a relatively positive threshold voltage, and thus the static power dissipation (leakage current) may be suppressed.
[0026] In some other embodiments in which the second semiconductor device **40** is a p-type TFT and the channel **41** is a p-channel, with the provision of the drain-side and source-side barrier layers **511**, **521** (each having a valence band edge energy that is lower than a valence band edge energy of the p-channel **41**), the threshold voltage of the p-type TFT **40** can be shifted from a relatively large positive value (without the drain-side and source-side barrier layers **511**, **521**) toward a relative small positive value or a negative value, thereby suppressing the static power dissipation of the second semiconductor device **40**. The first barrier height formed at the first interface is the difference between the valence band edge of the p-channel **41** and the valence band edge energy of the drain-side barrier layer **511**, and may be referred to as a first hole barrier height. The second barrier height formed at the second interface is the difference between the valence band edge energy of the p-channel **41** and the valence band edge energy of the source-side barrier layer **521**, and may be referred to as a second hole barrier height. In some embodiments, the first hole barrier height is greater than the second hole barrier height, and thus degradation of an on-state drain current may be prevented or alleviated (i.e., the on-state drain current (I.sub.on) of the second semiconductor device **40** is relatively large).
[0027] In some embodiments in which the second semiconductor device **40** is a p-type TFT, each of the drain-side and source-side barrier layers **511**, **521** may be independently selected from TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, Cs-doped nickel oxide (Cs:NiO.sub.x), or bathocuproine. In other words, each of the drain barrier material and the source barrier material may be independently selected from TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide (NiO.sub.x), or bathocuproine, and each of the first dopants and the second dopants may be independently selected from cesium (Cs), or other suitable dopants.
[0028] In some embodiments, each of the drain barrier material and the source barrier material may have a crystalline phase, an amorphous phase or a combination thereof. In some embodiments, the drain-side barrier layer **511** has a first thickness ranging from about 1 Å to about 300 Å. In some embodiments, the source-side barrier layer **521** has a second thickness ranging from about 1 Å to about 300 Å. The greater the thickness of each of the drain-side and source side barrier layers **511**, **521** (or the higher percentage of crystalline phase in each of the drain-side and source side barrier layers **511**, **521**) is, the greater the shift of the threshold voltage of the n-type TFT **40** (or the p-type TFT **40**) is.
[0029] In order to prevent or alleviate degradation of an on-state drain current, the energy band gap

of the drain-side barrier layer **511** is higher than the energy band gap of the source-side barrier layer **521**. In some embodiments, when the same dopants are applied to the drain-side and source-side barrier layers **511**, **521**, a dopant concentration in the drain-side barrier layer **511** is less than a dopant concentration in the source-side barrier layer **521** and/or the second thickness is less than the first thickness.

[0030] In some embodiments, the drain-side conductive metal oxide layer **512** (or the source-side conductive metal oxide layer **522**) is used to reduce a contact resistance between the channel **41** and the drain electrode **514** (or the source electrode **524**). As such, each of the drain-side and source-side conductive metal oxide layers **512**, **522** has an electrically conductivity which is lower than that of each of the drain and source electrodes **514**, **524**, and which is higher than that of the channel **41**. In addition, the electrically conductivity of each of the drain-side and source-side conductive metal oxide layers **512**, **522** is higher than that of each of the drain-side and source-side barrier layers **511**, **521**. In some embodiments, the drain-side conductive metal oxide layer **512** is in ohmic contact with the drain-side barrier layer **511** and the drain-side diffusion blocking layer **513**. In some embodiments, the source-side conductive metal oxide layer **522** is in ohmic contact with the source-side barrier layer **521** and the source-side diffusion blocking layer **523**. In some embodiments, each of the drain-side and source-side conductive metal oxide layers **512**, **522** may be independently made of indium gallium zinc oxide (IGZO), indium gallium oxide (IGO), indium zinc oxide (IZO), indium oxide, zinc oxide, zinc tin oxide (ZnSnO, ZTO), gallium zinc oxide (GZO), indium tin oxide (InSnO, ITO), fluorine-doped tin oxide (FTO), or other high conductive metal oxides having high donor (i.e., electrons) density. It is noted that, in comparison with the n-type metal oxide semiconductor for forming the channel **41** (such as the examples described in the previous paragraph), the metal oxides for forming the drain-side and source-side conductive metal oxide layers **512**, **522** has a relatively high donor (i.e., electrons) density. In other words, the conductivity of the metal oxides for forming the drain-side and source-side conductive metal oxide layers **512**, **522** is greater than the conductivity of the metal oxide semiconductor for forming the channel **41**. In some embodiments, each of the drain-side and source-side conductive metal oxide layers **512**, **522** may have a thickness ranging from about 5 nm to about 500 nm. The thicknesses of the drain-side and source-side conductive metal oxide layers **512**, **522** may be the same as or different from each other.

[0031] In some embodiments, each of the drain-side and source-side diffusion blocking layers **513**, **523** is provided for preventing metal atoms in a corresponding one of the drain and source electrodes **514**, **524** from diffusing into the IMD portion **31** of the interconnect layer M.sub.x, so that undesired current leakage may be eliminated. In some embodiments, each of the drain-side and source-side diffusion blocking layers **513**, **523** may include titanium nitride (TiN), tungsten carbon nitride (WCN), tungsten nitride (WN), tantalum nitride (TaN), etc. Other materials suitable for forming the drain-side and source-side diffusion blocking layers **513**, **523** are also within the contemplated scope of the present disclosure. In some embodiments, each of the drain-side and source-side diffusion blocking layers **513**, **523** may have a thickness ranging from about 0.5 Å to about 50 nm.

[0032] In some embodiments, the drain electrode **514** includes a first electrically conductive material, and the source electrode **524** includes a second electrically conductive material. The first electrically conductive material may be the same as or different from the second electrically conductive material. Each of the first and second electrically conductive materials may include tungsten (W), platinum (Pt), aluminum (Al), cobalt (Co), copper (Cu), titanium (Ti), tantalum (Ta), zirconium (Zr), hafnium (Hf), or alloys thereof. Other metallic materials suitable for forming the drain and source electrodes **514**, **524** are within the contemplated scope of the present disclosure.

[0033] In some embodiments, each of the drain and source electrodes **514**, **524** has an upper surface distal from the substrate **10**, a lower surface proximate to the surface **10**, and an interconnecting surface connecting the upper surface and the lower surface. In some embodiments, a drain-side film stack includes the drain-side barrier layer **511**, the drain-side conductive metal oxide layer **512** and the drain-side diffusion blocking layer **513** stacked on each other in such order, and a source-side film stack includes the source-side barrier layer **521**, the source-side conductive metal oxide layer **522** and the source-side diffusion blocking layer **523** stacked on each other in such order. In some embodiments, as shown in FIG. **1**, the drain-side film stack and the source-side film stack are respectively formed around the drain electrode **514** and the source electrode **524** such that the lower surface and the interconnecting surface of each of the drain and source electrodes **514**, **524** are covered by a respective one of the drain-side and source-side film stacks. In such case, not only the channel **41** is separated from each of the drain and source electrodes **514**, **524** by a respective one of the drain-side and source-side film stacks, but also the gate dielectric layer **42** is separated from each of the drain and source electrodes **514**, **524** by a respective one of the drain-side and source-side film stacks.

[0034] In some embodiments, as shown in FIG. **1**, the first confinement layer **44** is formed on a back surface **413** of the channel **41** opposite to a front surface **414** of the channel **41**. Each of the back surface **413** and the front surface **414** interconnects the drain-side surface **411** and the source-side surface **412**. The back surface **413** and the front surface **414** are respectively located distal from and proximate to the gate electrode **43**. An energy band gap of the first confinement layer **44** is greater than the energy band gap of the channel **41**. In some embodiments in which the second semiconductor device **40** is an n-type TFT and the channel **41** is an n-channel, a conduction band edge energy of the first confinement layer **44** is higher than the conduction band edge energy of the channel **41**. In some other embodiments in which the second semiconductor device **40** is a p-type TFT and the channel **41** is a p-channel, a valence band edge energy of the first confinement layer **44** is lower than the valence band edge energy of the channel **41**. Since a first carrier barrier height (i.e., the difference in the conduction band edge energy or the difference in the valence band edge energy) is formed at a first heterojunction between the channel **41** and the first confinement layer **44**, the majority carriers (electrons in an n-channel or holes in a p-channel) are confined to flow within the channel **41** by the first confinement layer **44**, thereby reducing a back-channel leakage current.

[0035] In some embodiments in which the second semiconductor device **40** is an n-type TFT, the first confinement layer **44** includes a first base material selected from gallium oxide, silicon oxide, or aluminum oxide. In some embodiments, the first confinement layer **44** has a first thickness ranging from about 1 Å to about 300 Å. In some embodiments, the first confinement layer **44** may further include first doping elements so as to adjust the energy band gap of the first confinement layer **44**. In some embodiments, the first doping elements doped in the first base material include indium, zinc or a combination thereof. For example, the first confinement layer **44**, which may be made of gallium oxide, zinc-doped gallium oxide or indium-doped gallium oxide, may be represented by a chemical formula of $(Ga_{x2}Q_{(1-x2)})O_{y2}$, where Q is Zn or In, x2 ranges from about 0.1 to about 1, and y2 ranges from about 0.1 to about 3. As a concentration of the first doping elements in the first confinement layer **44** increases, the energy band gap of the first confinement layer **44** decreases. The greater the energy band gap (or the greater thickness) of the first confinement layer **44** is, the more effective the reduction of the back-channel leakage current is. In some other embodiments in which the second semiconductor device **40** is a p-type TFT, the first confinement layer **44** may be selected from $TiSnO_3$, $Sn_5(PO_5)_2$, Ta_2SnO_6, Cs-doped nickel oxide ($Cs:NiO_x$), or bathocuproine. In other words, the first base material may be selected from $TiSnO_3$, $Sn_5(PO_5)_2$, Ta_2SnO_6, nickel oxide (NiO_x), or bathocuproine, and the first doping elements may include cesium (Cs), or other suitable dopants.

[0036] In some embodiments, when the second semiconductor device **40** is configured as a 1T FeRAM, where the gate dielectric layer **42** is made of a ferroelectric material such as HZO, the second semiconductor device **40** may further includes a first interface layer **45** formed between the gate dielectric layer **42** and the gate electrode **43**, and a second interface layer **46** formed between the gate dielectric layer **42** and the channel **41**. Each of the interface layers **45**, **46** is made of a high-k dielectric material, and is used for improving the endurance of the 1T FeRAM. In some embodiments, the first interface layer **45** may include titanium oxide, indium oxide, tantalum oxide, or combinations thereof. In some embodiments, the second interface layer **46** may include titanium oxide, indium oxide, or a combination thereof. In some embodiments, each of the interface layers **45**, **46** may have a thickness ranging from about 0.5 Å to about 100 nm.

[0037] FIG. **2** is a schematic sectional view illustrating the second semiconductor device **40** in accordance with some other embodiments. The second semiconductor device **40** shown in FIG. **2** has a structure similar to the second semiconductor device **40** shown in FIG. **1**, but further includes a second confinement layer **47** formed between the channel **41** and the gate dielectric layer **42**. In some embodiments, the second confinement layer **47** is formed on the front surface **414** of the channel **41** such that the first and second confinement layers **44**, **47** are respectively formed at two opposite sides of the channel **41**.

[0038] An energy band gap of the second confinement layer **47** is greater than the energy band gap of the channel **41**. In some embodiments in which the second semiconductor device **40** is an n-type TFT and the channel **41** is an n-channel, a conduction band edge energy of the second confinement layer **47** is higher than the conduction band edge energy of the channel **41**. In some other embodiments in which the second semiconductor device **40** is a p-type TFT and the channel **41** is a p-channel, a valence band edge energy of the second confinement layer **47** is lower than the valence band edge energy of the channel **41**. That is, a second carrier barrier height (i.e., the difference in the conduction band edge energy or the difference in the valence band edge energy) is formed at a second heterojunction between the channel **41** and the second confinement layer **47**. As shown in FIG. **2**, since the channel **41** (which has a relatively narrow energy band gap) is sandwiched between the first confinement layer **44** and the second confinement layer **47** (each having a relatively wide energy band gap), the majority carriers within the channel **41** and confined between the first and second confinement layers **44**, **47** may be considered as two-dimensional electron gas flowing in two dimensions, and the mobility of the majority carriers may be significantly enhanced, thereby increasing the on-state drain current (I.sub.on) of the second semiconductor device **40**.

[0039] In some embodiments in which the second semiconductor device **40** is an n-type TFT, the second confinement layer **47** includes a second base material selected from gallium oxide, silicon oxide, or aluminum oxide. In some embodiments, the second confinement layer **47** may further include second doping elements so as to adjust the energy band gap of the second confinement layer **47**. In some embodiments, the second doping elements doped in the second base material include indium, zinc or a combination thereof. For example, the second confinement layer **47**, which may be made of gallium oxide, zinc-doped gallium oxide or indium-doped gallium oxide, may be represented by a chemical formula of (Ga.sub.x3R.sub.(1-x3))O.sub.y3, where R is Zn or In, x3 ranges from about 0.1 to about 1, and y3 ranges from about 0.1 to about 3. As a concentration of the second doping elements in the second confinement layer **47** increases, the energy band gap of the second confinement layer **47** decreases. In some embodiments, the second confinement layer **47** has a first thickness ranging from about 1 Å to about 300 Å. In some embodiments, the energy band gap of the first confinement layer **44** is not less than the energy band gap of the second confinement layer **47**. Hence, when the same dopants are applied to the first and second confinement layers **44**, **47**, a dopant concentration in the first confinement layer **44** is less than a dopant concentration in the second confinement layer **47**, and/or the thickness of the first confinement layer **44** is not less than the thickness of the second confinement layer **47**. When the energy band gap of the second confinement layer **47** is greater than the energy band gap of the first confinement layer **44** or the thickness of the second confinement layer **47** is greater than the thickness of the first confinement layer **44**, an on-state drain current may be undesirably reduced.

In some other embodiments in which the second semiconductor device **40** is a p-type TFT, the second confinement layer **47** may be selected from TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, Cs-doped nickel oxide (Cs:NiO.sub.x), or bathocuproine. In other words, the second base material may be selected from TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide (NiO.sub.x), or bathocuproine, and the second doping elements may include cesium (Cs), or other suitable dopants.

[0040] FIG. **3** is a schematic sectional view illustrating the second semiconductor device **40** in accordance with some yet other embodiments. The second semiconductor device **40** shown in FIG. **3** has a structure similar to the second semiconductor device **40** shown in FIG. **2**, but the source-side barrier layer **521** shown in FIG. **2** is omitted. In such case, the channel **41** is instead in contact with the source-side conductive metal oxide layer **522**.

[0041] The source-side conductive metal oxide layer **522** disposed between the channel **41** and the source-side diffusion blocking layer **523** (or between the channel **41** and the source electrode **524**) may prevent formation of an interfacial metal oxide layer, which may be undesirably formed to include metal atoms from the source electrode **524** and oxygen atoms from the channel **41**, and which may be, for example, tungsten oxide, titanium oxide, aluminum oxide, and thus have a relatively high electrical resistance. In addition, due to the high donor (i.e., electrons) density of the source-side conductive metal oxide layer **522**, the source-side conductive metal oxide layer **522** may be in ohmic contact with the channel **41**. Therefore, degradation of an on-state drain current may be prevented or alleviated.

[0042] In some alternative embodiments, the semiconductor structure **1** and the second semiconductor device **40** may further include additional features, and/or some features present in the semiconductor structure **1** and the second semiconductor device **40** may be modified, replaced, or eliminated without departure from the spirit and scope of the present disclosure.

[0043] FIG. **4** is a flow diagram illustrating a method **6** for manufacturing the second semiconductor device **40** (e.g., the second semiconductor device **40** shown in FIG. **1**, **2** or **3**) in accordance with some embodiments. The method **6** may include steps S**01** to S**05**. FIGS. **5** to **11** illustrate schematic views of intermediate stages of the method **6** in accordance with some embodiments. Similar numerals from the above-mentioned embodiments have been used where appropriate, with some construction differences being indicated with different numerals. In some other embodiments, the method **6** may be used for manufacturing a plurality of the second semiconductor devices **40** simultaneously, and the number of the second semiconductor devices **40** may vary according to practical requirements.

[0044] Referring to FIG. **4** and the example illustrated in FIG. **5**, the method **6** begins at step S**01**, where the gate dielectric layer **42** is formed on the gate electrode **43** by atomic layer deposition (ALD), physical vapor deposition (PVD), chemical vapor deposition process (CVD), or other suitable deposition techniques. In some embodiments, prior to forming the gate dielectric layer **42**, the first interface layer **45** is formed on the gate electrode **43** using ALD, PVD, CVD or other suitable techniques. In some embodiments, a patterning process, which may include a photolithography process and an etching process following the photolithography process, may be performed so that each of the gate electrode **43**, the first interface layer **45** and the gate dielectric layer **42** may have a desired dimension in an X direction or in a Y direction that is transverse to the X direction.

[0045] Referring to FIG. **4** and the example illustrated in FIG. **6**, the method **6** proceeds to step S**02**, where the channel **41** and the first confinement layer **44** are sequentially formed on the first region **421** of the gate dielectric layer **42** by a suitable deposition process (such as the examples described in step S**01**) and a patterning process which may include a photolithography process and an etching process following the photolithography process. In some embodiments, prior to forming the channel **41**, the second interface layer **46** is formed on the first region **421** of the gate dielectric layer **42** by the above-mentioned processes. FIG. **6** is a schematic sectional view similar to that shown in FIG. **5**, but illustrating the structure after step S**02**.

[0046] In some embodiments, as shown in FIG. **6**, in the process for forming the second semiconductor device **40** shown in FIG. **2** or **3**, step S**02** may include forming the second interface layer **46**, the second confinement layer **47**, the channel **41** and the first confinement layer **44** sequentially on the first region **421** of the gate dielectric layer **42** by the above-mentioned processes. In some other embodiments, in the process for forming the second semiconductor device **40** shown in FIG. **1**, formation of the second confinement layer **47** may be omitted. That is, step S**02** may include forming the second interface layer **46**, the channel **41** and the first confinement layer **44** sequentially on the first region **421** of the gate dielectric layer **42** by the above-mentioned processes.

[0047] Referring to FIG. **4** and the example illustrated in FIG. **7**, the method **6** proceeds to step S**03**, where a dielectric layer **61** for forming the IMD portion **31** of the interconnect layer M.sub.x (see FIG. **1**, **2** or **3**) is formed on the structure obtained after step S**02** using a suitable deposition process (such as the examples described in step S**01**), followed by a planarization process, such as chemical mechanical polishing, to obtain a planar upper surface of the dielectric layer **61**. FIG. **7** is a schematic sectional view similar to that shown in FIG. **6**, but illustrating the structure after step S**03**.

[0048] As shown in FIG. **7**, in some embodiments, the dielectric layer **61** is formed on a stack of the second interface layer **46**, the second confinement layer **47**, the channel **41** and the first confinement layer **44**, and extends to cover the second and third regions **422**, **423** of the gate dielectric layer **42** opposite to the gate electrode **43**.

[0049] Referring to FIG. **4** and the example illustrated in FIG. **8**, the method **6** proceeds to step S**04**, where the dielectric layer **61** (see FIG. **7**) is patterned to have two openings **621**, **622** spaced apart from each other. Step S**04** is performed by, for example, a photolithography process and an etching process following the photolithography process. In addition, the two second regions **422** of the gate dielectric layer **42** are respectively exposed from the two openings **621**, **622**. The two openings **621**, **622** are respectively for forming the drain and source contact units **51**, **52** (see FIG. **1**, **2** or **3**) therein. FIG. **8** is a schematic sectional view similar to that shown in FIG. **7**, but illustrating the structure after step S**04**.

[0050] In some embodiments, as shown in FIG. **8**, two opposite side surfaces of each of the second interface layer **46**, the second confinement layer **47**, the channel **41** and the first confinement layer **44** are respectively exposed from the two openings **621**, **622**. That is, the drain-side surface **411** and the source-side surface **412** of the channel **41** are respectively exposed from the two openings **621**, **622**. After step S**04**, the patterned dielectric layer is denoted by the numeral **611**.

[0051] Referring to FIG. **4** and the example illustrated in FIGS. **2**, **3**, **9** to **11**, the method **6** proceeds to step S**05**, where the drain contact unit **51** is formed in the opening **621** (see FIG. **8**) and the source contact unit **52** is formed in the opening **622**, thereby obtaining the semiconductor device **40** shown in FIG. **1**, **2** or **3**. In some embodiments in which the drain and source contact units **51**, **52** have the same film stacking, as shown in FIG. **1** or **2**, the drain and source contact units **51**, **52** may be formed at the same time, while in some other embodiments in which the drain and source contact units **51**, **52** have different film stacking, as shown in FIG. **3**, the drain and source contact units **51**, **52** may be separately formed.

[0052] Specifically, in some embodiments, in the process for forming the drain and source contact units **51**, **52** shown in FIG. **1** or **2**, step S**05** may include sequentially depositing materials respectively for forming the drain-side and source-side barrier layers **511**, **521**, the drain-side and source-side conductive metal oxide layers **512**, **522**, the drain-side and source-side diffusion blocking layers **513**, **523**, and the drain and source electrodes **514**, **524** along an inner surface of each of the openings **621**, **622** over the patterned dielectric layer **611** to fill the openings **621**, **622** (see FIG. **8**) by suitable deposition processes, followed by performing a planarization process, such as chemical mechanical polishing, for a period of time until the patterned dielectric layer **611** is

exposed. Thereafter, the patterned dielectric layer **611** is formed into the IMD portion **31** of the interconnect layer M.sub.x.
[0053] In some other embodiments, in the process for forming the drain and source contact units **51**, **52** shown in FIG. **3**, step S**05** may include sub-steps shown in FIGS. **9** to **11** and **3**.
[0054] In the sub-step shown in FIG. **9**, a first masking layer **63** is partially formed on the structure obtained after step S**04** such that the opening **621** (see FIG. **8**) is covered by the first masking layer **63** and the opening **622** is exposed from the first masking layer **63**. In some embodiments, the first masking layer **63** may include an oxide, a nitride, a carbide, an oxynitride, an oxycarbide, a carbonitride, an oxycarbonitride, or combinations thereof. For example, the first masking layer **63** may be made of silicon oxide, aluminum oxide, hafnium oxide, zirconium oxide, silicon nitride, aluminum nitride, titanium nitride, silicon carbide, silicon oxycarbide, silicon oxynitride, silicon carbonitride, silicon oxycarbonitride, or combinations thereof. Other materials suitable for forming the first masking layer **63** are within the contemplated scope of the present disclosure.
[0055] In the sub-step shown in FIG. **10**, the source contact unit **52** (see also FIG. **3**) is formed after the first masking layer **63** is formed. In some embodiments, formation of the source contact unit **52** may include sequentially depositing materials respectively for forming the source-side conductive metal oxide layer **522**, the source-side diffusion blocking layer **523** and the source electrode **524** along the inner surface of the opening **622** (see FIG. **9**) to fill the opening **622** (see FIG. **9**) by suitable deposition processes; performing a planarization process, such as chemical mechanical polishing, for a period of time until the patterned dielectric layer **611** is exposed; and performing an etching process (e.g., a dry etching and/or a wet etching) to remove the first masking layer **63** (see FIG. **9**).
[0056] In the sub-step shown in FIG. **11**, after forming the source contact unit **52**, a second masking layer **64** is partially formed on the structure shown in FIG. **10** such that the source contact unit **52** is covered by the second masking layer **64** and the opening **621** is exposed from the second masking layer **64**. In some embodiments, possible materials suitable for forming the second masking layer **64** are similar to those for forming the first masking layer **63**, and thus the details thereof are omitted for the sake of brevity. Other materials suitable for forming the second masking layer **64** are within the contemplated scope of the present disclosure.
[0057] In the sub-step shown in FIG. **3**, the drain contact unit **51** (see also FIG. **3**) is formed after the second masking layer **64** is formed. In some embodiments, formation of the drain contact unit **51** may include sequentially depositing materials respectively for forming the drain-side barrier layer **511**, the drain-side conductive metal oxide layer **512**, the drain-side diffusion blocking layer **513** and the drain electrode **514** along the inner surface of the opening **621** (see FIG. **11**) to fill the opening **621** (see FIG. **11**) by suitable deposition processes, followed by performing a planarization process, such as chemical mechanical polishing, for a period of time until the patterned dielectric layer **611** is exposed. After the planarization process, the patterned dielectric layer **611** is formed into the IMD portion **31** of the interconnect layer M.sub.x. In some embodiments exemplified by FIGS. **3** and **9** to **11**, the drain contact unit **51** is formed after formation of the source contact unit **52**; while in some other embodiments, the drain contact unit **51** is formed before formation of the source contact unit **52**.
[0058] In some embodiments, some steps in the method **6** may be modified, replaced, or eliminated without departure from the spirit and scope of the present disclosure.
[0059] In summary, by selecting the materials of the drain-side and source-side barrier layers **511**, **521**, the threshold voltage (e.g., V.sub.t, or V.sub.t(initial), V.sub.t(PRG), V.sub.t(ERS), etc.) of the n-type TFT **40** may be shifted to a relatively positive value (e.g., shifted from about −0.02 V to about 0.26 V, or shifted from about −1 V to about 2 V), and the threshold voltage of the p-type TFT **40** may be shifted to a relatively negative value. Accordingly, current leakage may be suppressed and degradation of an on-state drain current (I.sub.on) may be alleviated. Furthermore, with the provision of the first and second confinement layers **44**, **47** which are respectively located at two

opposite sides of the channel **41** and which have a relatively wide energy band gap than that of the channel **41**, the mobility of the majority carriers within the channel **41** may be increased, thereby increasing the on-state drain current (I.sub.on) of the semiconductor device **40**. In addition, since the energy band gap of the first confinement layer **44** is relatively wide, the back-channel current leakage in the semiconductor device **40** may be further reduced.

[0060] In accordance with some embodiments of the present disclosure, a method for manufacturing a semiconductor structure includes: forming a gate electrode; forming a gate dielectric layer over the gate electrode; forming a channel over the gate dielectric layer opposite to the gate electrode, the channel including a semiconductor material; forming a first confinement layer on the channel opposite to the gate electrode, an energy band gap of the first confinement layer being greater than an energy band gap of the channel; and forming a drain electrode and a source electrode which are connected to the channel, the drain electrode and the source electrode being spaced apart from each other and each including an electrically conductive material.

[0061] In accordance with some embodiments of the present disclosure, the semiconductor material has an n-type conductivity, and a conduction band edge energy of the first confinement layer is higher than a conduction band edge energy of the channel.

[0062] In accordance with some embodiments of the present disclosure, the semiconductor material has a p-type conductivity, and a valence band edge energy of the first confinement layer is lower than a valence band edge energy of the channel.

[0063] In accordance with some embodiments of the present disclosure, the method further includes forming a second confinement layer between the channel and the gate dielectric layer. An energy band gap of the second confinement layer is greater than the energy band gap of the channel.

[0064] In accordance with some embodiments of the present disclosure, the energy band gap of the first confinement layer is not less than the energy band gap of the second confinement layer.

[0065] In accordance with some embodiments of the present disclosure, a thickness of the first confinement layer is not less than a thickness of the second confinement layer.

[0066] In accordance with some embodiments of the present disclosure, the first confinement layer includes a first base material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine, and the second confinement layer includes a second base material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine.

[0067] In accordance with some embodiments of the present disclosure, the first confinement layer further includes first doping elements, and the second confinement layer further includes second doping elements.

[0068] In accordance with some embodiments of the present disclosure, the first doping elements include indium, zinc, cesium, or combinations thereof, and the second doping elements includes indium, zinc, cesium, or combinations thereof.

[0069] In accordance with some embodiments of the present disclosure, a method for manufacturing a semiconductor structure includes: forming a gate electrode; forming a gate dielectric layer over the gate electrode; forming a channel over the gate dielectric layer opposite to the gate electrode, the channel including a semiconductor material; forming a confinement layer on the channel opposite to the gate electrode, an energy band gap of the confinement layer being greater than an energy band gap of the channel; forming a drain-side barrier layer in contact with the channel so as to form a first barrier height at an interface between the drain-side barrier layer and the channel, an energy band gap of the drain-side barrier layer being greater than the energy band gap of the channel; and forming a drain electrode and a source electrode, the drain electrode being connected to the channel through the drain-side barrier layer, the source electrode being connected to the channel, the drain electrode and the source electrode being spaced apart from each other and each including an electrically conductive material.

[0070] In accordance with some embodiments of the present disclosure, the method further includes forming a source-side barrier layer between the source electrode and the channel so as to form a second barrier height at an interface between the source-side barrier layer and the channel. An energy band gap of the source-side barrier layer is greater than the energy band gap of the channel.

[0071] In accordance with some embodiments of the present disclosure, the first barrier height is greater than the second barrier height.

[0072] In accordance with some embodiments of the present disclosure, a thickness of the drain-side barrier layer is not less than a thickness of the source-side barrier layer.

[0073] In accordance with some embodiments of the present disclosure, the drain-side barrier layer includes a drain barrier material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine, and the source-side barrier layer includes a source barrier material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine.

[0074] In accordance with some embodiments of the present disclosure, the drain-side barrier layer further includes first dopants selected from indium, zinc, cesium, or combinations thereof, the source-side barrier layer further includes second dopants selected from indium, zinc, cesium, or combinations thereof, and the energy band gap of the drain-side barrier layer is greater than the energy band gap of the source-side barrier layer.

[0075] In accordance with some embodiments of the present disclosure, the method further includes: forming a drain-side conductive metal oxide layer between the drain electrode and the drain-side barrier layer; and forming a source-side conductive metal oxide layer which is disposed between the source electrode and the channel and which is in ohmic contact with the channel.

[0076] In accordance with some embodiments of the present disclosure, the drain-side conductive metal oxide layer includes indium gallium zinc oxide, indium gallium oxide, indium zinc oxide, indium oxide, zinc oxide, zinc tin oxide, gallium zinc oxide, indium tin oxide, fluorine-doped tin oxide, or combinations thereof, and the source-side conductive metal oxide layer includes indium gallium zinc oxide, indium gallium oxide, indium zinc oxide, indium oxide, zinc oxide, zinc tin oxide, gallium zinc oxide, indium tin oxide, fluorine-doped tin oxide, or combinations thereof.

[0077] In accordance with some embodiments of the present disclosure, a semiconductor structure includes: a gate electrode; a gate dielectric layer disposed over gate electrode, the gate dielectric layer including a ferroelectric material; a channel disposed over the gate dielectric layer, the channel including a semiconductor material; a drain contact unit and a source contact unit connected to the channel and spaced apart from each other; and a first confinement layer disposed on the channel opposite to the gate electrode, an energy band gap of the first confinement layer being greater than an energy band gap of the channel.

[0078] In accordance with some embodiments of the present disclosure, the semiconductor structure further includes a second confinement layer disposed between the channel and the gate dielectric layer. An energy band gap of the second confinement layer is greater than the energy band gap of the channel.

[0079] In accordance with some embodiments of the present disclosure, the drain contact unit includes: a drain-side barrier layer disposed on the channel, an energy band gap of the drain-side barrier layer being greater than the energy band gap of the channel; a drain-side conductive metal oxide layer disposed on the drain-side barrier layer opposite to the channel; and a drain electrode disposed on the drain-side conductive metal oxide layer opposite to the drain-side barrier layer, the drain electrode including a first electrically conductive material. The source contact unit includes a source electrode which includes a second electrically conductive material, and a source-side conductive metal oxide layer which is disposed between the source electrode and the channel and which is in ohmic contact with the channel.

[0080] In accordance with some embodiments of the present disclosure, a method for manufacturing a semiconductor structure includes: forming a gate electrode; forming a gate dielectric layer over gate electrode, the gate dielectric layer including a ferroelectric material; forming a channel over the gate dielectric layer, the channel including a semiconductor material; forming a confinement layer over the channel opposite to the gate electrode, an energy band gap of the confinement layer being greater than an energy band gap of the channel; and forming a drain contact unit connected to the channel; and forming a source contact unit connected to the channel, the source contact unit being spaced apart from the drain contact unit.

[0081] In accordance with some embodiments of the present disclosure, the drain contact unit and the source contact unit are separately formed.

[0082] In accordance with some embodiments of the present disclosure, formation of the drain contact unit includes forming a drain-side barrier layer on the channel, forming a drain-side conductive metal oxide layer on the drain-side barrier layer opposite to the channel, and forming a drain electrode on the drain-side conductive metal oxide layer opposite to the drain-side barrier layer. Formation of the source contact unit includes forming a source-side conductive metal oxide layer on and in ohmic contact with the channel, and forming a source electrode on the source-side conductive metal oxide layer opposite to the channel. An energy band gap of the drain-side barrier layer is greater than the energy band gap of the channel. Each of the drain electrode and the source electrode includes an electrically conductive material.

[0083] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes or structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for manufacturing a semiconductor structure, comprising: forming a gate electrode; forming a gate dielectric layer over the gate electrode; forming a channel over the gate dielectric layer opposite to the gate electrode, the channel including a semiconductor material; forming a first confinement layer on the channel opposite to the gate electrode, an energy band gap of the first confinement layer being greater than an energy band gap of the channel; and forming a drain electrode and a source electrode which are connected to the channel, the drain electrode and the source electrode being spaced apart from each other and each including an electrically conductive material.

2. The method as claimed in claim 1, wherein the semiconductor material has an n-type conductivity, and a conduction band edge energy of the first confinement layer is higher than a conduction band edge energy of the channel.

3. The method as claimed in claim 1, wherein the semiconductor material has a p-type conductivity, and a valence band edge energy of the first confinement layer is lower than a valence band edge energy of the channel.

4. The method as claimed in claim 1, further comprising forming a second confinement layer between the channel and the gate dielectric layer, an energy band gap of the second confinement layer being greater than the energy band gap of the channel.

5. The method as claimed in claim 4, wherein the energy band gap of the first confinement layer is not less than the energy band gap of the second confinement layer.

6. The method as claimed in claim 4, wherein a thickness of the first confinement layer is not less than a thickness of the second confinement layer.
7. The method as claimed in claim 4, wherein the first confinement layer includes a first base material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine, and the second confinement layer includes a second base material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine.
8. The method as claimed in claim 7, wherein the first confinement layer further includes first doping elements, and the second confinement layer further includes second doping elements.
9. The method as claimed in claim 8, wherein the first doping elements include indium, zinc, cesium, or combinations thereof, and the second doping elements includes indium, zinc, cesium, or combinations thereof.
10. A method for manufacturing a semiconductor structure, comprising: forming a gate electrode; forming a gate dielectric layer over the gate electrode; forming a channel over the gate dielectric layer opposite to the gate electrode, the channel including a semiconductor material; forming a confinement layer on the channel opposite to the gate electrode, an energy band gap of the confinement layer being greater than an energy band gap of the channel; forming a drain-side barrier layer in contact with the channel so as to form a first barrier height at an interface between the drain-side barrier layer and the channel, an energy band gap of the drain-side barrier layer being greater than the energy band gap of the channel; and forming a drain electrode and a source electrode, the drain electrode being connected to the channel through the drain-side barrier layer, the source electrode being connected to the channel, the drain electrode and the source electrode being spaced apart from each other and each including an electrically conductive material.
11. The method as claimed in claim 10, further comprising forming a source-side barrier layer between the source electrode and the channel so as to form a second barrier height at an interface between the source-side barrier layer and the channel, an energy band gap of the source-side barrier layer being greater than the energy band gap of the channel.
12. The method as claimed in claim 11, wherein the first barrier height is greater than the second barrier height.
13. The method as claimed in claim 11, wherein a thickness of the drain-side barrier layer is not less than a thickness of the source-side barrier layer.
14. The method as claimed in claim 11, wherein the drain-side barrier layer includes a drain barrier material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine, and the source-side barrier layer includes a source barrier material selected from gallium oxide, silicon oxide, aluminum oxide, TiSnO.sub.3, Sn.sub.5(PO.sub.5).sub.2, Ta.sub.2SnO.sub.6, nickel oxide, or bathocuproine.
15. The method as claimed in claim 14, wherein the drain-side barrier layer further includes first dopants selected from indium, zinc, cesium, or combinations thereof, the source-side barrier layer further includes second dopants selected from indium, zinc, cesium, or combinations thereof, and the energy band gap of the drain-side barrier layer is greater than the energy band gap of the source-side barrier layer.
16. The method as claimed in claim 10, further comprising: forming a drain-side conductive metal oxide layer between the drain electrode and the drain-side barrier layer; and forming a source-side conductive metal oxide layer which is disposed between the source electrode and the channel and which is in ohmic contact with the channel.
17. The method as claimed in claim 16, wherein the drain-side conductive metal oxide layer includes indium gallium zinc oxide, indium gallium oxide, indium zinc oxide, indium oxide, zinc oxide, zinc tin oxide, gallium zinc oxide, indium tin oxide, fluorine-doped tin oxide, or

combinations thereof, and the source-side conductive metal oxide layer includes indium gallium zinc oxide, indium gallium oxide, indium zinc oxide, indium oxide, zinc oxide, zinc tin oxide, gallium zinc oxide, indium tin oxide, fluorine-doped tin oxide, or combinations thereof.

18. A semiconductor structure, comprising: a gate electrode; a gate dielectric layer disposed over gate electrode, the gate dielectric layer including a ferroelectric material; a channel disposed over the gate dielectric layer, the channel including a semiconductor material; a drain contact unit and a source contact unit connected to the channel and spaced apart from each other; and a first confinement layer disposed on the channel opposite to the gate electrode, an energy band gap of the first confinement layer being greater than an energy band gap of the channel.

19. The semiconductor structure as claimed in claim 18, further comprising a second confinement layer disposed between the channel and the gate dielectric layer, an energy band gap of the second confinement layer being greater than the energy band gap of the channel.

20. The semiconductor structure as claimed in claim 18, wherein the drain contact unit includes a drain-side barrier layer disposed on the channel, an energy band gap of the drain-side barrier layer being greater than the energy band gap of the channel, a drain-side conductive metal oxide layer disposed on the drain-side barrier layer opposite to the channel, and a drain electrode disposed on the drain-side conductive metal oxide layer opposite to the drain-side barrier layer, the drain electrode including a first electrically conductive material, and the source contact unit includes a source electrode which includes a second electrically conductive material, and a source-side conductive metal oxide layer which is disposed between the source electrode and the channel and which is in ohmic contact with the channel.